

Feature spreading

OVA · 2026 · Andrew Lamont

6 August 2026

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1 Features spread

Tuparí¹ (tpr, Tupian, Brazil)

¹ [tu.pa.'ri.]

(1) **[nasal] spreading in Tuparí** (Singerman, 2016:454-455, 463-464, 467)

- | | | |
|--------------------|------------------|---------------------|
| a. /o-aot-a/ | [.wao.ra.] | 'I leave' |
| b. /o-m-aot-a/ | [.õ.mãõ.řã.] | 'make me leave' |
| c. /o-m-aot-a-p/ | [.õ.mãõ.řãm.] | 'making me leave' |
| d. /o-m-aot-a-t/ | [.õ.mãõ.řãn.] | 'made me leave' |
| e. /ko-roro/ | [.ko.ro.ro.] | 'eat, drink-NMLZ' |
| f. /nẽ-roro/ | [.nẽ.řõ.řõ.] | 'do, make-NMLZ' |
| g. /te-we-m-ato-a/ | [.te.we.mã.toa.] | 'bathe one another' |

Directional rule-based approach (oversimplification)

- [−nasal] → [+nasal] / [+nasal] _ || ⇒
self-feeding: output becomes context
/omaota/ → omaota → omãota → omãõta → omãõřã → omãõřã
- same idea with autosegmental representation

Iteration problem for HS (Pater et al., 2007)

- Replacing a constraint's violation does not improve

(2) /o-m-aot-a/ → [.õ.mãõ.řã.]

/o-m-aot-a/	MAX(link)	*[+nasal][−nasal]	DEP(link)
● a. omaota		1	L
b. obaota	W 1	L	
☹ c. omãota		1	1
d. õmaota		1	1

1.1 Bread crumb approaches

Common tactic

- assign * to every possible target
à la Hansel laying down a trail of bread crumbs
- spreading feature follows trail, eats *
compare $\text{PARSE}(\sigma)$ for footing

spreading as alignment (Bickmore, 1996; Jurgec, 2011)

(3) $\text{ALIGN}([\text{nasal}], \text{R}, \text{word}, \text{R}, \text{segment})$ (Piggott, 1996:148)

For every [nasal], assign one violation for every segment that intervenes between its R edge and the R edge of the nearest word.

“No segments follow nasal spans within words.”

(4) /o-m-aot-a/ → [õ.mãõ.řã.]

		MAX(link)	ALIGN(...)	DEP(link)
	/o-m-aot-a/			
	a. omaota		W 4	L
	b. obaota	W 1	L	L
→	c. omãota		3	1
	d. õmaota		W 4	1
→	e. omãota		W 3	L
→	f. omãõta		2	1
→	g. omãõta		W 2	L
→	h. omãõta		1	1
→	i. omãõta		W 1	L
→	j. omãõtä			1
→	k. omãõtä			

problem: ALIGN constraints cannot account for blocking

(5) **Voiceless obstruent onsets block nasalization in Tuparí** (Singerman, 2016:462, 465, 467-468, 470)

- | | | |
|--------------------|-------------------|----------------------|
| a. /ãpe/ | [.ã.pe.] | ‘comb’ |
| b. /m-ato-a/ | [.mã.toa.] | ‘bathe someone else’ |
| c. /e-m-aoros-a-p/ | [.ẽ.mãõ.řõ.sapˀ.] | ‘making you arrive’ |
| d. /māk-a/ | [.mã.ka.] | ‘send, hand over’ |
| e. /e-õ-ʔet-a-t/ | [.ẽõ.ʔe.ratˀ.] | ‘made you sleep’ |

- position in morpheme matters – we’re simplifying
- [.mã.toa.] incurs three violations of ALIGN(...)
- cannot be resolved by spreading
weird prediction: truncate to right of blockers
[.mã.toa.] * [.mã.]
weird prediction: epenthesis only before [nasal]
/te.pẽan-aoros-a/ → [.te.pẽ.ãñ.ʔa.o.ro.sa.] (Singerman, 2018:396)
weird prediction: prosodic word truncates
/omaota/ → * [⟨.om.⟩ ⟨.a.o.ta.⟩]
Side note: generalizations in Pruitt (2022) may be derived by early vs. late prosodic word parsing?
computational complaints (Eisner, 1997; Biró, 2003; Lamont, 2021)

Could we change the target?

- align to nearest obstruent?
no – emãõtosap has 1 violation
→ emãõ̃tosap has 2 violations
not harmonically improving

spreading as sharing (McCarthy, 2010; Mullin, 2011)

(6) SHARE(F)

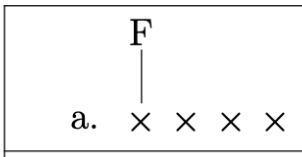
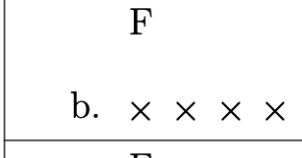
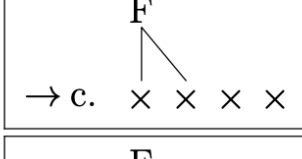
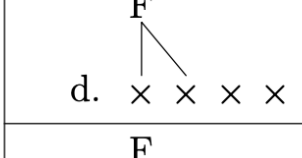
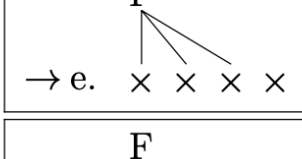
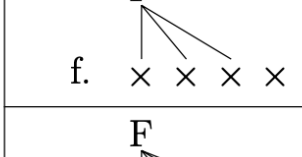
Assign one violation mark for every pair of adjacent elements that are not linked to the same token of F.

- assigns violations independently of presence of F
- same issues of preferring short words

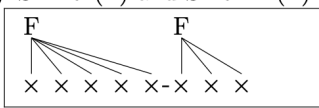
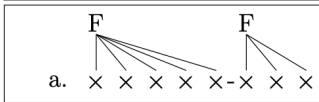
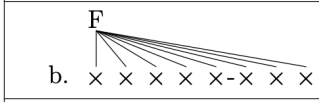
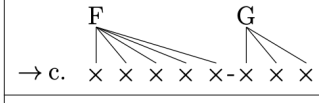
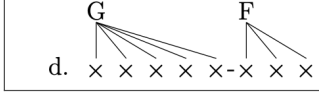
Bad prediction: a count effect (Lamont, 2022)

- rule-based models are finite-state (Johnson, 1972; Koskeniemi, 1983; Kaplan and Kay, 1994)
- determining whether $m > n$ is not

(7) (5.38) SHARE(F) motivates autosegmental spreading

	SHARE (F)	DEP (link)
a. 	W 3	L
b. 	W 3	L
→ c. 	2	1
d. 	W 2	L
→ e. 	1	1
f. 	W 1	L
→ g. 		1

(8) (5.41) SHARE(F) and SPREAD(F) produce count effects (unattested)

	*F...F	DEP (link)	IDENT	SHARE (F)	SPREAD (F)
a. 	W 1		L	1	L 6
b. 		W 1	L	L	L 7
→ c. 			1	3	4
d. 			1	W 6	W 2

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